Anomaly Detection in Wastewater Treatment Plants Using Unsupervised Machine Learning Algorithms.

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Abstract

The timely detection of unusual wastewater influent contaminants or treatment defaults can significantly mitigate the release of sub-quality treated sewage effluent. Hence, protecting the marine life and the ecosystem from adverse impacts resulting from water pollution. This paper studies and recommends the best anomaly detection algorithm in the wastewater treatment industry. The study was conducted using several unsupervised Machine Learning (ML) algorithms, precisely the Isolation Forest Algorithm (IFA), Fast Minimum Covariance Determinant algorithm (FMCD), Local Outlier Factor algorithm (LOF), and One-Class Support Vector Machines (OCSVM) algorithm. Along with studying the effect of applying Principle Components Analysis on improving anomaly detection performance. In addition, it compared two strategies for applying anomaly detection. The anomaly detection results showed that the FMCD is the most robust algorithm for identifying anomalies in various dimensions, scoring a recall and F1Score using a univariate anomaly detection strategy exceeding 0.90 in four wastewater parameters. Similarly, it scored using a multivariate anomaly detection strategy recall and an F1Score of 0.76. The study found that multivariate anomaly detection and removal strategy is preferred for modeling tasks. However, a univariate anomaly detection strategy is recommended for plant operations to avoid the high dimensionality curse, which deteriorates the model detection performance. The recommended anomaly detection algorithm and strategy can be deployed in an integrated framework of remote sensing and ML to detect wastewater anomalies at wastewater treatment plants, leading to an effective wastewater treatment control that can contribute to preventing environmental pollution.

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1. Introduction

Wastewater treatment is fundamental to protecting public health, preserving the environment, removing organic matter, solids, nutrients, and disease-causing organisms before wastewater reuse or disposal. Specific government Treated Sewage Effluent (TSE) parameters must be met before returning the TSE to irrigation networks or disposing it to water bodies. Releasing untreated wastewater to water bodies leads to a drastic environmental impact due to microorganisms' consumption of oxygen (Miao et al., 2021; Yaqub et al., 2020). Wastewater microorganisms can lead to an inhabitable water body due to depleting dissolved oxygen and infectious pathogens (Rama Karri, 2021).

Generated raw wastewater is highly complicated in terms of its chemical composition due to the biological nature and limitless components that can be found (Metcalf & Eddy, 2015; Cheng T.F. et al., 2020). However, only critical components are of concern that must be well assessed to design suitable treatment processes. Such components must be closely monitored to evaluate the treatment operation's reliability and efficiency.

One of these components is the Chemical Oxygen Demand (COD). It is a measure of oxygen equivalent for determining the biodegradable organic matter that can be oxidized chemically using a potent oxidization agent such as potassium dichromate (Metcalf & Eddy, 2015; Cheng T.F. et al., 2020). Another component is the Biochemical Oxygen Demand (BOD5). It measures oxygen equivalent to determine the biodegradable organic matter that can be oxidized biologically through microorganisms (Metcalf & Eddy, 2015). In other words, it represents how much organic materials must be removed. Therefore, detecting abnormal observations from these features is critical since it directly impacts the biological treatment process. Failure to detect such events will result in elevated

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amounts of biodegradable organics being released to water bodies without removal, leading to depletion of oxygen and septic conditions in the water body.

A further critical wastewater parameter is Total Kjeldahl Nitrogen (TKN), which measures total organic and ammonia nitrogen in wastewater (Metcalf & Eddy, 2015). It can be described as a measure of the nitrogenous oxygen demand required for the survival of microorganisms. Furthermore, Phosphorus is also an important parameter that is needed to be monitored. It appears in many forms in wastewater. Phosphorus can be found in the form of orthophosphate, polyphosphates, and organic phosphate, which is essential for biological organisms' growth.

The wastewater treatment process is nonlinear and requires monitoring and controlling of various operating parameters, wastewater characteristics, and chemical reactions (Wang D. et al., 2021). Due to that, operators with adequate expert knowledge are needed to be able to run such a plant. In such situations, Machine Learning (ML) models can be utilized.

Machine Learning (ML) can model complex nonlinear relationships without explicitly describing the treatment process by mathematical or chemical equations due to its ability to extract relationships from data. Unlike the legacy models (Wang D. et al., 2021; Wang J. K. et al., 2020), it has great potential to explore a new understanding of wastewater behavior that was difficult to see. ML integration with wastewater treatment processes will aid operators as a guidance tool to improve the treatment process, reduce operational costs (Moldovan, 2019), optimize plant performance (Icke et al., 2020; Arismendy et al., 2020), and improve preserving the environment.

Cheng T. et al. (2019) proposed WWTP wastewater influent monitoring model to detect anomalies, using a Kernel Principal Component Analysis (KPCA) and OCSVM. The authors have compared several kernels and PCs. The results showed that a polynomial Kernel with ten PCs and OCSVM achieves a perfect anomaly detection of value one in terms of True Positive, AUC, & F1Score through identifying all anomalies in the dataset.

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Dairi et al. (2019) developed an anomaly detection model for influent conditions to act promptly, minimize downstream treatment failures, and disturb effluent quality. The model was developed by capturing temporal features using an integrated RNN-Restricted Boltzmann Machines algorithm with OCSVM. The modeling results were compared against several other algorithms, which showed that the developed anomaly detection model achieves 0.98 AUC & 0.99 F1Score.

Icke et al. (2020) proposed an anomaly detection model for the nitrate, ammonium, and aeration treatment. In addition to detecting instrument anomalies using quantile regression, combined with an Artificial Neural Network. The authors' case study promptly detected the ammonium sensor's failure using the developed algorithm, which led to an unnecessarily increase in the aeration flow.

Mamandipoor et al. (2020) developed an influent ammonia sensor's fault detection classification model. The model was developed using the LSTM algorithm trained on over 12 features containing over 435,000 data points. The dataset was collected in one year at a one-minute frequency achieving 92% fault detection recall.

Due to the high complexity, the uncertainty of wastewater behavior (Wand D. H. et al., 2020), and sensitivity to the weather conditions, the WWTPs are likely to be impacted by abnormal wastewater influent contamination or loading that may exceed the plant treatment capacity, leading to a failure in the treatment process and potential risk on the environment (Metcalf & Eddy, 2015). For instance, an uncommon event can result from chemical contamination, equipment failure, sensor measurement drift, or operator control default. Although these suspicious events are a minority, they drastically impact the wastewater treatment process and environment, especially if not detected promptly. Such failures may lead to tank overflow, treatment cost penalties imposed by authorities, or emergency disposal to water bodies, which will destroy the marine environment in the affected area (Icke et al., 2020). Therefore, an intelligent automatic abnormal events detection system must be developed for WWTPs.

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The main contribution of this paper is to fill the literature gap in comparing different unsupervised learning anomaly detection algorithms in the wastewater treatment industry and recommending the best algorithm and anomaly detection strategy for wastewater. The recommended anomaly detection algorithm and strategy can be integrated into a wastewater anomaly detection framework using the Internet of Things (IoT) and ML.

The paper studies the following: i) to compare four anomaly detection algorithms' performance against abnormal wastewater contaminations; ii) to explore univariate and multivariate anomaly detection strategy performance; iii) to study the effect of applying Principal Component Analysis (PCA) on anomaly detection performance; iv) to discuss findings and recommend best wastewater anomaly detection policy.

This paper is organized as follows. In section 2, the method and modeling algorithms are presented. Section 3 analyzes the dataset used in the study. Section 4 delivers the modeling results and discussion. Finally, section 5 concludes.

2. Methods & Modeling Algorithms

Principle Component Analysis (PCA) and four anomaly detection algorithms were applied to identify the abnormal events and examine their performance. The techniques are the Isolation Forest algorithm (IFA), Fast Minimum Covariance Determinant algorithm (FMCD), Local Outlier Factor algorithm (LOC), and One-Class Support Vector Machines algorithm (OCSVM).

2.1. Principle Component Analysis

PCA is an unsupervised learning algorithm for dimension reduction and data compression (Abba et al., 2020). Let $x_1,\ldots..,x_n \in \mathrm{D}$, where $n \in \mathbb{N}$ and having t features. First, the data are centered by standardization to have a mean equal to zero and a standard deviation equal to one. The vector defines the new space direction u_1 and to be a unit vector where $u_1^T u_1 = 1$. The data x_n is rotated by multiplying it with a rotation matrix u_1^T to project it to the new space. The projected data variance is presented in (1).

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$$V = \frac{1}{N}\sum_{n=1}^{N}\{u_1^T x_n - u_1^T \bar{x}\}^2 = u_1^T S u_1 \quad (1)$$

The covariance matrix S in (2) is computed to evaluate whether the dataset contains redundant information.

$$S = \frac{1}{N}\sum_{n=1}^{N}(x_n - \bar{x})(x_n - \bar{x})^T \quad (2)$$

To increase the projected data variance $u_1^T S u_1$ from the origin with respect to u_1 and to avoid having space vector $u_1 = \infty$, u_1 needs to be constrained subject to having $u_1^T u_1 = 1$. As such, the Lagrange multiplier λ_1 is introduced (Younes et al., 2021). Hence, the projection variance maximization is shown in (3).

$$V = u_1^T S u_1 + \lambda_1(1 - u_1^T u_1) \quad (3)$$

By taking the derivative of the above equation with respect to u_1, resulting $S u_1 = \lambda_1 u_1$. By multiplying both sides by u_1^T, the equation becomes $u_1^T S u_1 = \lambda_1$.

The above equations can be interpreted as the maximum variance can be obtained at the eigenvector having the highest eigenvalue λ_1 in the covariance matrix S. The eigenvector with the highest eigenvalue is called the first principal component, which holds the most information.

2.2. Anomaly Detection Algorithms

An anomaly event is an abnormal event that deviates from the majority. Anomalies are divided into two types which are global and local. Global anomalies are easier to detect as they are far from the whole dataset range. However, local anomalies are more difficult to identify since they are within the dataset range but away from surrounding neighbor observations (Alghushairy et al., 2020). Anomaly detection identifies patterns within the dataset that differ from the expected behavior (Inoue et al., 2017). It is an application to identify suspicious observations and rare events resulting from off-spec influent or effluent, sensors measurement failure, equipment failure, operator default, abnormal influent loadings such as rainfall events, or the existence of toxic components (Inoue et al., 2017). Their values are different from regular observations. Practical,

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timely monitoring of treatment operation parameters assists operators during abnormal events in promptly adjusting the operation controllers to mitigate any chances of failure and maintain a stable, high-quality treatment process. Similarly, anomalies not removed before data modeling tasks will likely adversely affect the model inference capabilities.

Anomaly Detection applications are divided into supervised & unsupervised learning methods. Supervised learning anomaly detection algorithms require labeled data to identify abnormal observations. On the other hand, unsupervised learning anomaly detection algorithms such as LOF & IFA build their model solely from historical data without labeling.

The Local Outlier Factor algorithm (LOF) is unsupervised learning density-based anomaly detection algorithm (Breunig et al., 2000). Consider random nearest neighbor $k \in \mathbb{N}$. Followed by calculating k-distance d_k for every point $o_n \in D \in \mathbb{R}^n$. Next, number of neighbor points q is found if point $N_k(p) = q \in D\backslash\{p\} | d(p,q) \leq d_k(p)$. We then calculate reachability distance $r_{d_k}(p,o_n) = \max\{d(p,o_n), d_k(o_n)\}$. Accordingly, local reachability density $lrd_k(p)$ and local outlier factor $LOF_k(o)$ are calculated using (4) & (5). An LOF value ≤ 1 is an inlier, and it is an outlier if it is much higher than 1 (Ma, 2013). Detailed explanations and equations are available in (Alghushairy et al., 2020; Breunig et al., 2000).

$$lrd_k(p) = \frac{\|N_k(p)\|}{\sum_{o \in N_k(p)} r_{d_k}(p,o_n)} \tag{4}$$

$$LOF_k(o) = \frac{\sum_{o \in N_k(p)} \frac{lrd_k(o_n)}{lrd_k(p)}}{\|N_k(p)\|} \tag{5}$$

The Fast Minimum Covariance Determinant (FMCD) algorithm is a multivariate location and scatters estimator using a covariance matrix. Let $x_1,.....,x_n \in \mathrm{D}$, where $n \in \mathbb{N}$ and having t features. Taking subset $h \in \mathrm{D}$ of size ϵ, weighted average $\hat{\mu}_{RMCD} = \sum_i^h x_i/h$ and $\mathrm{C_{step}}$ parameter

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$C = (1/h)\sum_i^h (x_i - \hat{\mu}_{RMCD})(x_i - \hat{\mu}_{RMCD})^T$ are computed. To capture most of the dataset observations, compute robust distance $RD(x)$ using (6) and sort in descending order π. Then, we create a new subset h' of size ϵ using the smallest $RD(x)$ and recompute $\hat{\mu}_{RMCD}$ & C till convergence or a given specific threshold. The smallest covariance determinant subsamples are the inliers. Algorithm explanations and equations are available in (Sarda et al., 2021; Rousseeuw & Hubert, 2018; Verardi & Dehon, 2010; Grosklos & Theiler, 2015; Rousseeuw & Driessen, 1999).

$$RD(x) = \sqrt{(x_n - \hat{\mu}_{RMCD})^T C^{-1} (x_n - \hat{\mu}_{RMCD})} \quad (6)$$

The Isolation Forest algorithm (IFA) is an unsupervised learning anomaly detection algorithm (Liu et al., 2008). Let dataset subset $\psi(x_1,...,x_n) \in \mathrm{D}(x_1,...,x_N)$. Forest of trees were grown with a splitting attribute q and split value s where nodes n split if $q < s$ till reaching maximum depth, or all data ended at the terminal node. Each point $p_n \in \psi$ anomaly score A is computed by measuring path length $h(x)$ divided by a normalization factor $c(n)$ as shown in (7) & (8). Where H is the harmonic number. If the anomaly score returns a value close to one, the observation is likely an anomaly, and it is a regular observation if the anomaly score is close to 0.5. Algorithm details are available in (Liu et al., 2008).

$$c(n) = 2H(n-1) - (2(n-1)/n) \quad (7)$$

$$A(x, n) = 2^{-\frac{E(h(x))}{c(n)}} \quad (8)$$

The One-Class Support Vector Machines algorithm (OCSVM) is an extension of the Support Vector Machines (SVM) algorithm for unsupervised learning anomaly detection. Let $x_1,.....,x_n \in \mathrm{D}$, where $n \in \mathbb{N}$. Dataset D is projected via kernel map ⍰ into feature space F to maximize the data separation from origin through optimizing (9).

$$min_{w,\,p,\,c} \quad \frac{1}{2}\|w\|^2 + \frac{1}{\nu N}\sum_i^N \varepsilon_i - p \quad (9)$$

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Where w are weights, v is a regularization term, and p is the offset. To transform the data from the current dimension Z to a higher dimension G, let $\phi: Z \rightarrow G$ be the kernel map. Then, using the hyperplane equation shown in (10), the decision function returns +1 in the most captured data region or -1 (outlier) elsewhere (Schölkopf et al., 1999). Where α_i is Lagrange multiplier, and $K(z,z_i)$ is the kernel function. Detailed explanations and equations are available in (Cheng T. et al., 2019; Schölkopf et al., 1999).

$$f(z) = sign\{(w.\phi(z)) - p\} = sign\left\{\sum_{i=1}^{N} \alpha_i K(z,z_i) - p\right\} \quad (10)$$

2.3. Model Performance Evaluation

The performance of the unsupervised learning algorithms was validated against manual labeling based on four metrics. The first metric is Tue Positive (TP), defined as the number of observations correctly identified and classified as positive cases. The second metric is the True Negative (TN), defined as the number of observations correctly identified and classified as negative cases. The third metric is the False Positive (FP), defined as the number of negative observations misclassified as positive cases. The last metric is the False Negative (FN), defined as the number of positive observations misclassified as negative cases.

From the above metrics, three essential metrics can be further derived, which are $Precision\ (P) = TP/(TP + FP)$, $Recall\ (R) = TP/(TP + FN)$, and $F1Score = 2PR/(P + R)$.

The anomaly detection pseudo code is presented in Table 1 below. The integrated framework of WWTP and the machine learning model is presented in Fig.1.

Algorithm: Anomaly Detection (IFA, FMCD, LOF, OCSVM)

Inputs: *D* – Dataset, *v* – Contamination, *s* – Subset (if required)

Output: Anomaly observations

1: Drop unnecessary features
2: Replace missing values with arithmetic mean
3: Normalize data with their mean and standard deviation
4: Apply PCA #Skip line 4 if not required
5: **For** *i*=1 to D_{width} **do**
#Only for univariate strategy. Otherwise, skip lines 5 & 9

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6:	Anomalies = Anomaly Detection algorithm *(D, v, s)*
7:	Anomalies index = Where (Anomalies= -1) *#stop here if the task is only to detect anomalies; otherwise, continue till line 10.*
8:	D' (Anomalies index)=NaN (Not a Number)
9:	**End for**
10:	Drop NaN values from the dataset

Table 1. Anomaly detection pseudo code **(Double column table)**

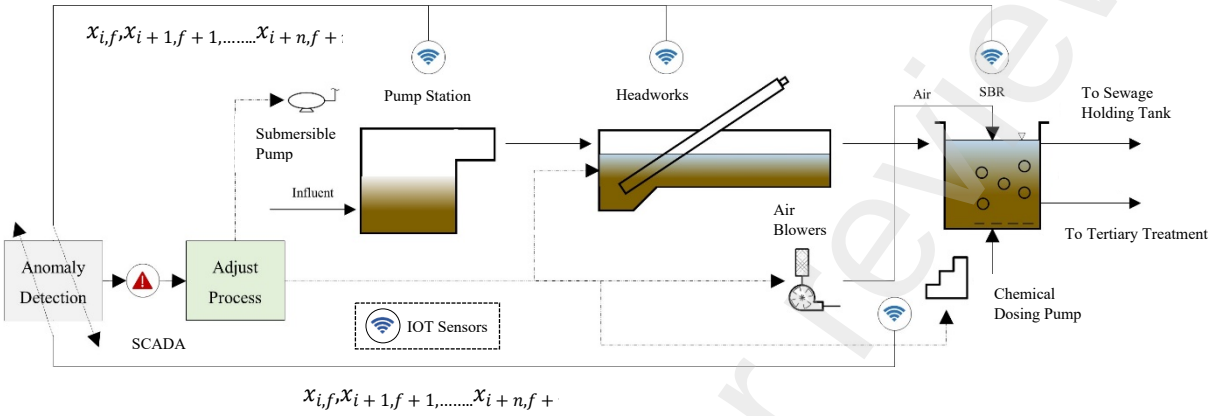


Fig. 1. The proposed anomaly detection framework is applied to the preliminary and secondary treatment process. IoT sensors remotely capture the wastewater characteristics. The anomaly detection model detects abnormal events to assist the plant operators in adjusting the treatment process. **(Double column figure - color)**

3. Data Analysis

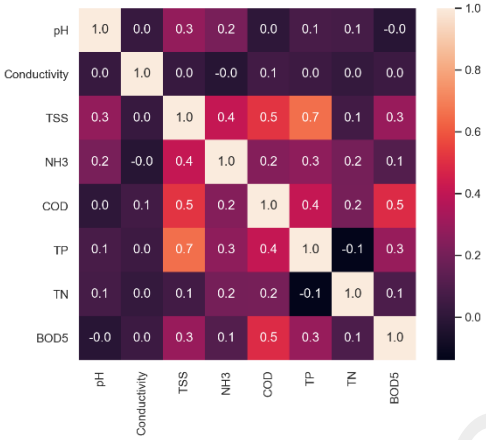
This section introduces the dataset used for anomaly detection using unsupervised ML algorithms. The case study was done on Askar WWTP, located in the southern governorate of the Kingdom of Bahrain. The data have been collected from the Ministry of Works, Sanitary Engineering Operation & Maintenance Directorate.

Askar WWTP is a 1,000 m^3/day capacity plant located in Madinat Khalifa, southern Bahrain. It was built in 2019 and operated in 2020 to serve the Riyad Askar housing project consisting of 541 villas. The plant was designed based on Sequencing Batch Reactor (SBR) Activated Sludge Process to provide a tertiary class TSE. The dataset covers approximately two years of readings collected between April 2020 and February 2022. It contains 30 features with 673 daily readings each. The features are divided into 12 influent features, ten aeration stage features, and eight effluent features. The dataset does not include any readings on holidays and specifically on Fridays. The merged cells were unmerged, and the categorical values were deleted. Since the features are on different scales, the

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features were normalized. The supplementary materials include the dataset features description (Table A.1), dataset statistics (Table A.2), and visualization (Fig. A.1).

A correlation matrix 

is plotted in Fig. 2. Surprisingly, there is a moderate correlation of 0.5 between COD & BOD, while it should be higher. This is justified due to the existence of anomalies. A higher correlation of 0.7 is found between Total Suspended Solids (TSS) & Total Phosphorus (TP). It can be interpreted that the more TSS there, the more TP is expected.

Fig. 2. Askar WWTP Correlation Matrix **(Single column figure – color)**

4. Results & Discussion

The study was conducted as follows. First, the four unsupervised anomaly detection algorithms were applied following a univariate anomaly detection strategy. Then, the study was repeated following a multivariate anomaly detection strategy. In the end, the PCA was applied to observe the effect on anomaly detection performance. The v hyperparameter for OCSVM and contamination hyperparameter for other algorithms varies depending on the feature anomalies.

In the univariate anomaly detection case, Table 2 shows that the OCSVM algorithm with Radial Basis Function (RBF) kernel performs poorly by scoring low recall performance. It can be seen in Fig. 3, especially pH, NH_3, and TN features, that the OCSVM algorithm tends to identify too many points as anomalies. The algorithm divides the dense region into several regions separated by outlier

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instances. It tends to fit hyperplanes to capture most instances, identifying regular observations in the middle of high-density regions as both normal and anomaly points, while it should only recognize them all as regular points. Another observation is the BOD5 figure; the algorithm did not capture apparent anomalies away from the high dense cluster. On the other hand, it identified observations in the middle of the high-density cluster as anomalies. It is justified that the algorithm fitted a separate hyperplane for these obvious abnormal observations, which blocked the algorithm from identifying them as anomalies.

The contrary was observed with the LOF algorithm. It failed to identify apparent anomalies when applied to pH and TN. The algorithm could not identify a true anomaly at all. It scored zero precision and recall in these two parameters. For instance, specific points were identified as anomalies in the TN feature, although they have relatively acceptable local density and are surrounded by other points. While significantly away points with both low local & global density failed to be identified. A similar incident was observed against TSS. In addition, LOF had relatively acceptable performance against other parameters. It scored the best performance compared to other algorithms against the TP feature only.

On the other hand, IFA had a moderate performance. It only performed poorly against TP. The algorithm scored the highest results against TSS and TN, slightly exceeding FMCD performance. IFA scored 0.63 precision and 0.62 recall against TSS. In comparison, it scored 0.89 precision and 0.82 recall against TN. It should be highlighted that the maximum sample hyperparameter significantly affects the algorithm performance. The mentioned hyperparameter specifies the number of observations drawn to train each tree. Tuning such hyperparameter can result in performance improvement that reaches magnitudes. The results shown in Table 2 are based on the 256-maximum number of samples.

Lastly, FMCD outperformed all algorithms. It achieved superior performance in four of eight features scoring a precision and recall exceeding 0.90. In contrast, it scored comparable performance to the best results obtained by IFA against TSS & TN. Most of the apparent anomalies were detected.

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Figure A.2 in supplementary materials shows NH_3, TN, and BOD5 anomaly detection results in time-series presentation.

Algorithm (Contamination %)		Anomaly Detection Results							
		pH (0.016)	**Conductivity (0.015)**	**TSS (0.086)**	**NH_3 (0.048)**	**COD (0.016)**	**TP (0.062)**	**TN (0.074)**	**BOD5 (0.018)**
Actual Anomalies		**11**	**10**	**58**	**32**	**11**	**42**	**50**	**12**
IFA	Detected	8	11	**57**	26	11	34	**46**	13
	Precision	0.73	0.64	**0.63**	0.85	0.73	0.44	**0.89**	0.85
	Recall	0.73	0.70	**0.62**	0.69	0.73	0.36	**0.82**	0.92
	F1-Score	0.73	0.67	**0.63**	0.76	0.73	0.39	**0.85**	0.88
FMCD	Detected	**10**	11	54	**31**	**11**	42	41	**12**
	Precision	**1.00**	0.64	0.61	**0.97**	**0.91**	0.67	0.90	**0.92**
	Recall	**0.91**	0.70	0.57	**0.94**	**0.91**	0.67	0.74	**0.92**
	F1-Score	**0.95**	0.67	0.59	**0.95**	**0.91**	0.67	0.81	**0.92**
LOF	Detected	4	10	50	20	11	**39**	42	13
	Precision	0	0.50	0.62	1.00	0.73	**0.97**	0	0.85
	Recall	0	0.50	0.53	0.62	0.73	**0.90**	0	0.92
	F1-Score	0	0.50	0.57	0.77	0.73	**0.95**	0	0.88
OCSVM	Detected	204	**9**	64	258	63	96	129	35
	Precision	0.02	**0.78**	0.27	0.09	0.10	0.12	0.16	0.09
	Recall	0.45	**0.70**	0.29	0.69	0.55	0.29	0.42	0.25
	F1-Score	0.05	**0.74**	0.28	0.15	0.16	0.17	0.23	0.13

Table 2. Askar influent features dataset – anomaly detection results **(Double column table)**

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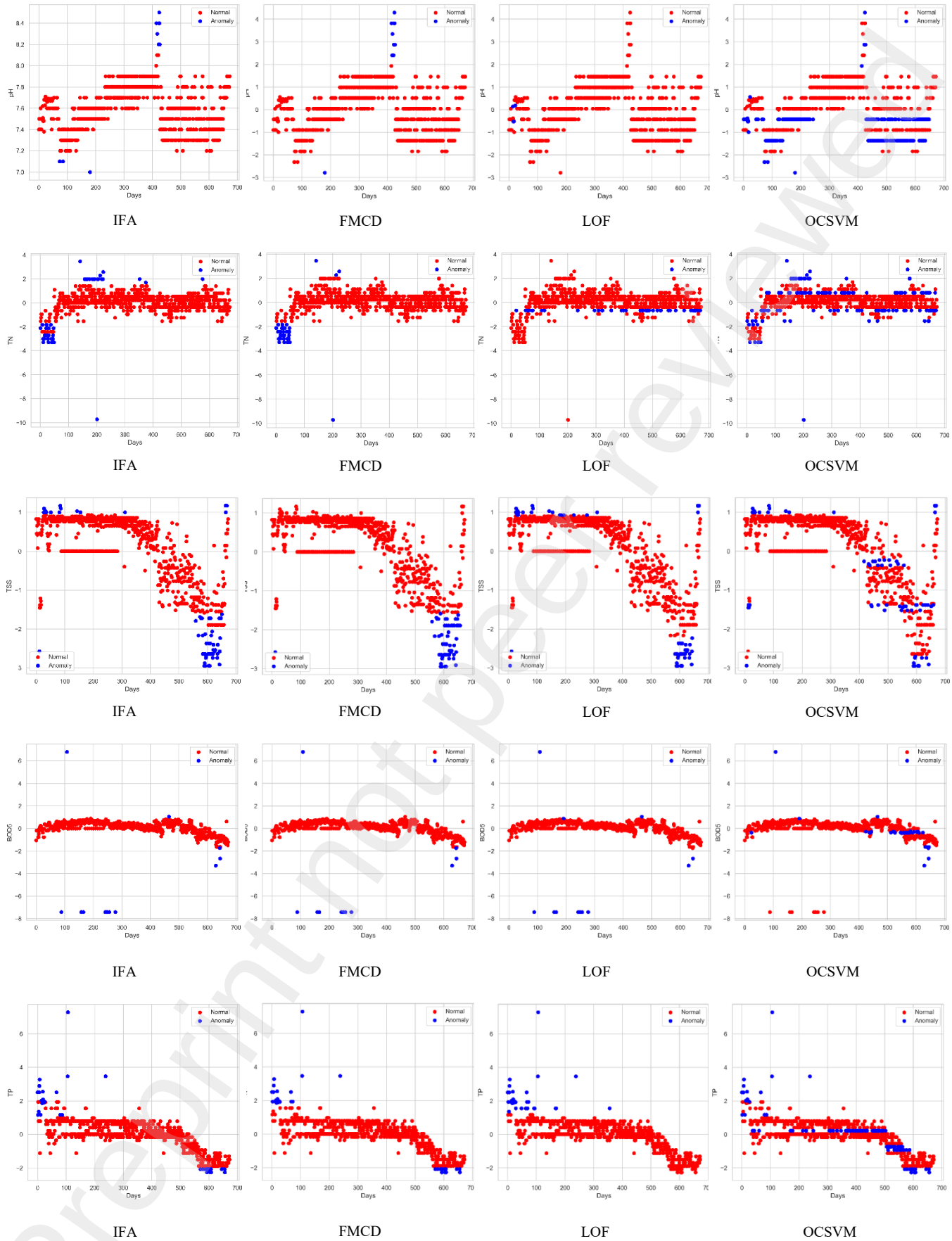


Fig. 3. Anomalies are highlighted in blue for influent wastewater features using four unsupervised ML algorithms. Each column shows the anomaly detection results using different algorithms indicated at the bottom of the column. The rows show pH, TN, TSS, BOD5, and TP anomalies. **(double column figure – color)**

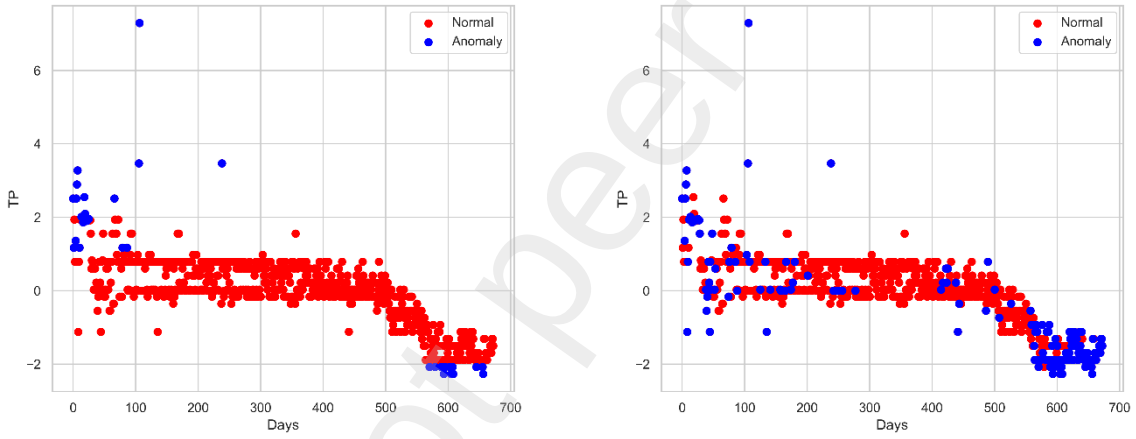
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In the multivariate anomaly detection case, most points detected by IFA univariate detection were also detected by multivariate detection. Figure 4.1 compares TP feature IFA anomaly detection using both strategies.

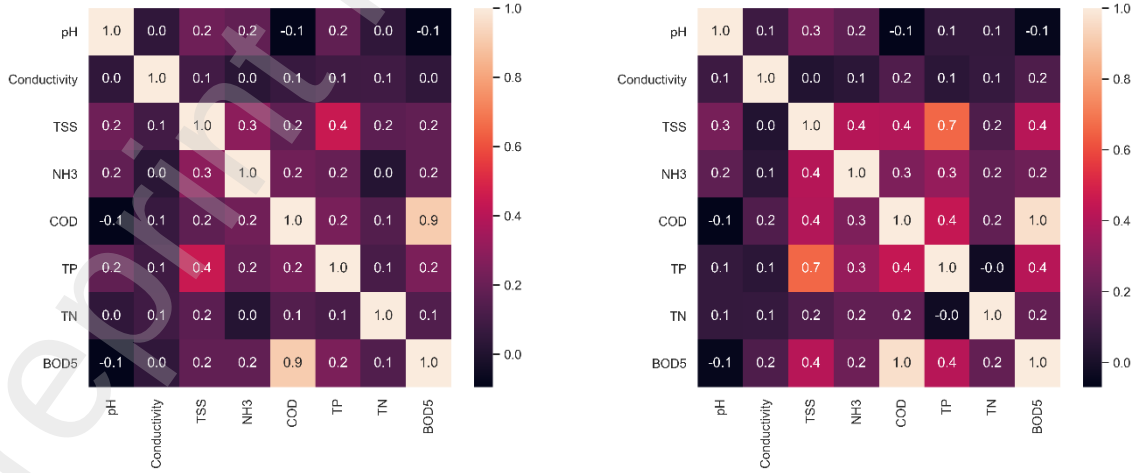
A correlation matrix was developed to visualize the performance difference. The correlation matrix in Fig. 4.2 shows that the multivariate anomaly detection strategy removed more true positive anomalies than the univariate anomaly detection strategy, resulting in an enhanced feature correlation. The correlation between TP & TSS increased from 0.4 in Fig. 4.2.a to 0.7 in Fig. 4.2.b. Similarly, BOD5 and COD correlation increased from 0.9 to 1.0. This shows the significant impact of anomaly detection strategy on future modeling tasks.

(4.1)



(a) Univariate Detection (b) Multivariate Detection

(4.2)



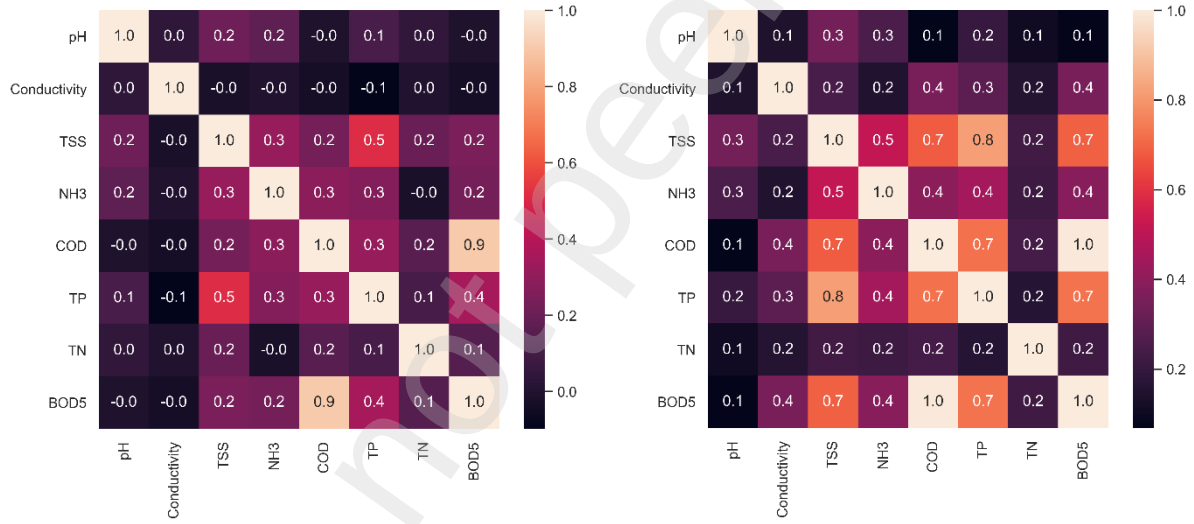
(a) Univariate Detection (b) Multivariate Detection

Fig. 4. TP feature – Comparison between IFA univariate and multivariate anomaly detection strategies **(Single column figure – color)**

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The same examination was applied to the FMCD algorithm in Fig. 5. It shows significantly different results between both anomaly detection strategies. The correlation between TP & BOD5 increased from 0.4 in Fig. 5.a to 0.7 in Fig. 5.b. Similarly, between COD & TP and between TSS & TP. The correlation between COD & BOD5 also improved from 0.9 to a perfect correlation of one. By comparing Fig. 5.a and Fig. 5.b, the correlation between TSS & BOD5 and COD & TSS almost tripled.

Accordingly, correlation matrices of both IFA multivariate anomaly detection in Fig. 4.2.b and FMCD multivariate anomaly detection Fig. 5.b clearly show that FMCD has overall significant features correlation improvement after removing outliers, which is considered a significant improvement for data modeling in case it was used as outlier removal.



(a) Univariate Detection (b) Multivariate Detection

Fig. 5.FMCD – Correlation comparison of anomaly detection strategies. **(Single column figure – color)**

To complete the study, LOF was revisited. A multivariate anomaly detection strategy was applied. Surprisingly, the opposite was found. The points not identified when the algorithm was applied using a univariate strategy were identified using multivariate anomaly detection. Figure 6 compares pH and TN features using both strategies.

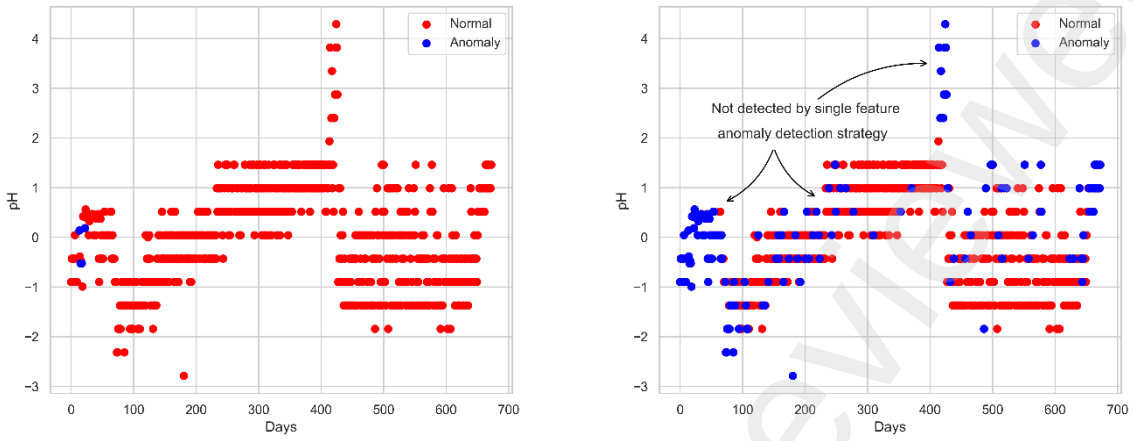
A correlation matrix was plotted in Fig. 6.3. It is noticeable that feature correlation in both cases is improved. Unlike the other algorithms, the univariate anomaly detection strategy had a

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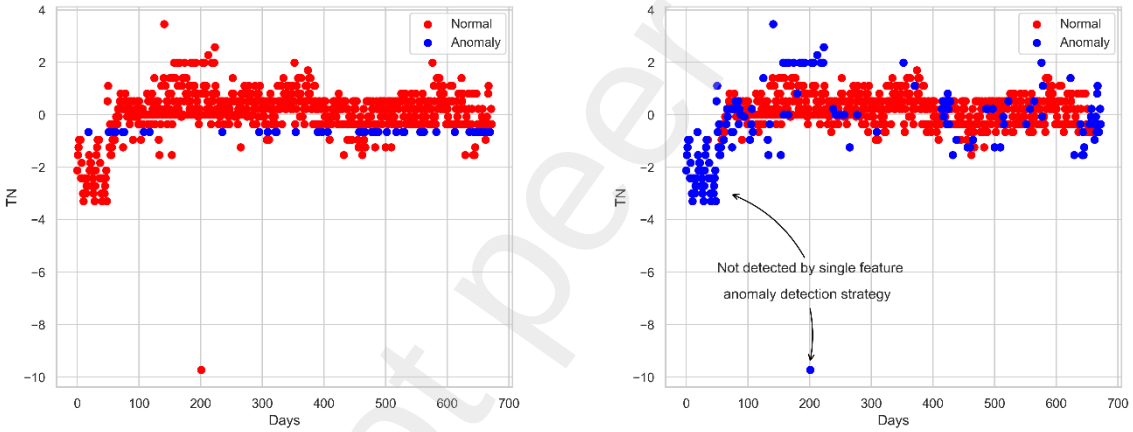
correlation that was not far from the group features strategy. The result conflicts with the other algorithms' behavior when changing the anomaly detection strategy.

(6.1)



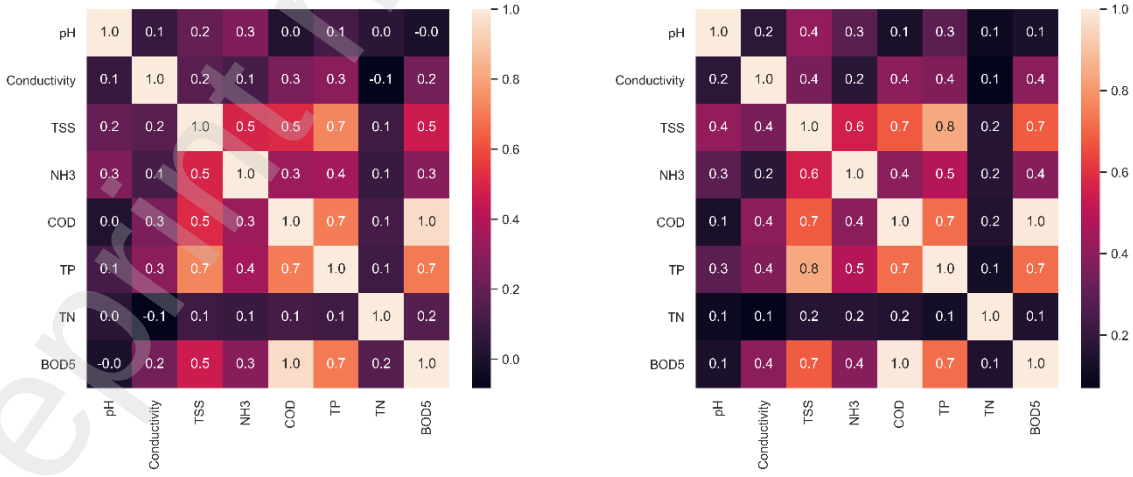
(a) Univariate Detection (b) Multivariate Detection

(6.2)



(a) Univariate Detection (b) Multivariate Detection

(6.3)



(a) Univariate Detection (b) Multivariate Detection

Fig. 6. LOF – pH (6.1), TN (6.2) Features and correlation - Comparison between univariate and multivariate anomaly detection. **(Double column figure – color)**

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To decide the best algorithm performance, the dataset features were manually labeled. A total of 163 anomaly observations were identified. The unsupervised algorithms' predictions were compared against the manual labeling. A contamination hyperparameter of 0.242 was used since there are 163 anomalies out of 673 observations. As shown earlier, a multivariate anomaly detection strategy was used to avoid drawbacks.

The results in Table 3 show that IFA performs the worst. It only detected 108 true positives out of 163, scoring 0.66 recall. In contrast, all other algorithms achieved a comparable performance. FMCD detected 124 true positives out of 163, scoring 0.76 recall. Although LOF did not perform well in univariate anomaly detection, as shown in Table 3, it achieved good performance detecting 117 true positives, scoring 0.72 recall. However, the LOF drawback is by being not able to be applied to predict anomalies from a new dataset. It must be re-trained every time a new observation is included (McKinnon, 2020). Hence, it is not a practical algorithm. Similarly, despite OCSVM suffering in univariate cases, it detected 128 true positive scoring 0.79 recall. Table 3 summarizes the findings.

	TP	**TN**	**FN**
IFA	108	455	55
	Precision	**Recall**	**F1-Score**
Outlier	0.66	0.66	0.66
Inlier	0.89	0.89	0.89

	TP	**TN**	**FN**
FMCD	124	471	39
	Precision	**Recall**	**F1-Score**
Outlier	0.76	0.76	0.76
Inlier	0.92	0.92	0.92

	TP	**TN**	**FN**
LOF	117	465	46
	Precision	**Recall**	**F1-Score**
Outlier	0.72	0.72	0.72
Inlier	0.91	0.91	0.91

	TP	**TN**	**FN**
OCSVM	128	476	35
	Precision	**Recall**	**F1-Score**
Outlier	0.79	0.79	0.79
Inlier	0.93	0.93	0.93

Table 3. Anomaly detection report summary (Contamination = 0.242) **(Double column table)**

Another case was implemented by transforming the dataset features using PCA. The PCA components number was kept equal to the features number. The PCA-based IFA algorithm results showed an 8% performance improvement compared to conventional IFA algorithm performance. The recall increased from 0.66 to 0.72, making it match LOF algorithms results. On the other hand, the other PCA-based anomaly detection algorithms deteriorated and could not achieve enhanced performance.

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What needs to be highlighted is that anomaly detection strategy does impact the algorithm performance. The results shown prove two points. The first point is to select an anomaly detection algorithm that matches the dataset's nature. The more the algorithm anomaly detection mechanism fits the feature behavior, the better detection results can be obtained. The second point is that it is not only about selecting an appropriate anomaly detection algorithm but also selecting a suitable detection strategy.

In the univariate anomaly detection case, FMCD outperformed other algorithms. The algorithm tends to identify extreme high or low observations as anomalies rather than observations in the middle of the high-density clusters. The algorithm methodology justifies this. It tries to fit an ellipse to capture most of the data. Such a concept fits the time-series nature of wastewater readings that tend to take an elliptical shape. Thus, anomalies are easily identified.

Unlike OCSVM, the algorithm captures several clusters to identify an abnormal event, even between two dense clusters, as shown in Fig. 3. Such an event might be an anomaly in another field, but not wastewater treatment. Similarly, the LOF algorithm anomaly detection process is based on the k-nearest neighbor. Such an algorithm is robust in identifying local outliers between regular readings. However, the algorithm suffers from fluctuating performance. This is clearly illustrated with pH and TN features. Lastly, IFA acceptably performed in both univariate and multivariate anomaly detection. Therefore, from a theory point of view and the above analysis, all findings point towards FMCD as the recommended algorithm for anomaly detection for wastewater parameters.

As a result of the above discussion, the recommendation is divided into two scenarios. From practicality, redundancy, and operations scenario, wastewater parameters anomaly detection is recommended to be monitored individually. Thus, the univariate feature anomaly detection strategy is recommended to be followed due to being more effective in anomaly identification and mitigating the chances of strong dimensionality influence on performance. In this case, FMCD is the recommended anomaly detection algorithm. The model would be more robust if only checking anomalies on individual feature dimensions. Hence, it will assess the plant operator in detecting true anomalies

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effectively and promptly. In practice, the algorithm will operate in a completely unsupervised way. Thus, the hyperparameters must be adjusted to fit each feature in the related WWTP.

On the contrary, for data modeling & outlier removal scenarios, the recommended strategy is the multivariate features anomaly detection strategy using the FMCD algorithm. The algorithm is robust against identifying anomalies in various dimensions. The dataset is examined on multiple dimensions simultaneously to detect outliers on the dataset level, not the univariate level, that may mislead the dataset modeling stage for prediction tasks. An outlier point in low dimensional representation might not be an outlier on higher dimensional representation and vice versa. Hence, if such outliers in the whole dataset are well-identified and removed, the dataset correlation will dramatically improve, as seen with FCMD multivariate anomaly detection correlation matrix compared to the dataset's original state in Fig. 2. Such dataset preprocessing shows that data analysis and proper treatment are necessary to improve the modeling results.

Furthermore, it is recommended for outlier removal only to be conducted on features that will be used during the modeling. Any features that will not be used during modeling shall be removed from the dataset before anomaly detection and removal to avoid affecting the detection performance, especially if the strategy followed is multivariate anomaly detection. This is because more features will elevate the anomaly detection dimensionality. Hence, more likely to change the detection results.

5. Conclusion

This paper demonstrated the importance of anomaly detection in wastewater treatment plants to avoid treatment process failures or penalties. The paper presented the anomaly detection results and concluded that from a practical & operation point of view, a univariate anomaly detection strategy should be followed using FMCD algorithms. However, for anomaly removal, the multivariate anomaly detection strategy with the FMCD algorithm is the most robust algorithm for detecting anomalies in various dimension datasets.

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The anomaly detection acts as a decision-supporting tool to assist the WWTP operators in promptly receiving the required information. Hence, it enables effective monitoring of wastewater parameters, timely detection of abnormal contamination, adjusting the operation controllers, taking corrective actions to minimize any chances of failure, mitigating environmental risks of treatment defaults, and maintaining a stable treatment process that meets the effluent quality parameters.

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